

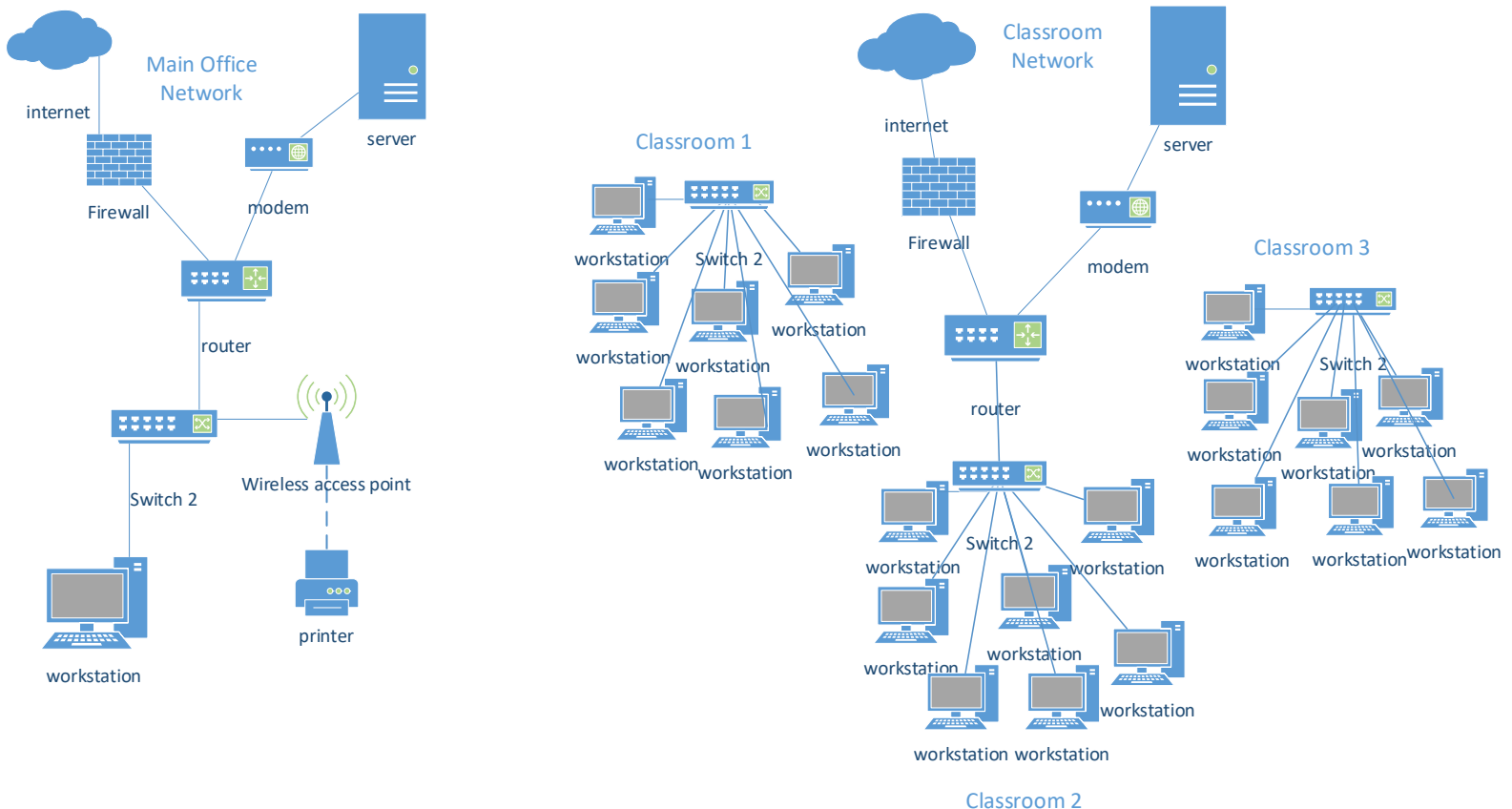
Network Topology Error Detection and Analysis of Solutions
Business Proposal for Thunderhawk Institute
Jessie Drake
IT510: Systems Analysis and Design
March 15, 2026

Executive Summary

Thunderhawk Institute is experiencing a network issue that involves upwards of twenty-one (21) workstations in three separate classrooms all connected to one switch for internet connectivity. Since small groups frequent the school to learn about computer programs, it is essential there is a continuous and reliable network to sustain the student's education. The recommendation for providing lasting internet connection is to invest in wireless access points for each classroom to reduce the need for complicated cabling and increase the ability to add different types of technology like mobile devices to the network if desired in the future.

Systems Evaluation

The network is comprised of one switch to encompass the main office and all three classrooms of computers. This requires significant cabling and organizing to maintain. It also places a lot of strain on the single switch and can result in significant downtime if it fails. The reliance on one switch can also prevent flexibility when the school desires to incorporate different types of technologies like printers or projectors that can require wireless internet connectivity.



Solution 1

Adding in an additional switch for each classroom and routing each of those switches to a central switch can be a possible solution to the current network errors. For technical efficiency, network speeds and reliability of the components of the network can increase. It can also save time if one switch fails, the IT team can easily detect and resolve the failing component and minimize disturbance to the network as a whole. In terms of technical feasibility, this solution will take some initial setup to restructure the network into separate switches for each of the three classrooms. For economic and operational feasibility, the additional switches will require an initial upfront cost but will ultimately result in lower maintenance costs and less downtime when issues occur.

Solution 2

Another solution to the issue of the single switch can involve the addition of wireless access points in each classroom. This will allow for increased technical efficiency as there will be significantly less cabling involved. Procedural efficiency will also improve as there can be additional equipment like a printer or mobile devices easily connected to improve flexibility and scalability for different needs in the future. In terms of technical feasibility, there will be an upfront cost of the wireless access points, but considering economic and operational feasibility, it will increase scalability and ease of use for students, teachers, administrators, and IT teams when issues arise.

Solution 3

If no corrective measures are taken, when there is a network connectivity problem, the IT team will spend longer than needed determining where the issue is originating from. This will result in the entire network being down while the issue is being resolved. Another consequence is the difficulty in navigating the increase demand or need for newer technology. Since the one switch is already handling the entirety of the network, incorporating more technology can overload the switch and prevent scalability.

Final Recommendation

The preferred solution for Thunderhawk Institute is solution 2. With the increasing need for wireless connectivity to add devices and increase technical capabilities within the institution, the addition of wireless access points makes the most sense in terms of economic, financial, and operational feasibility. IT teams will be able to discover issues within the network quickly and devise solutions without the entire network being affected for long periods of time.